

Conduct ABC, FSN and HML analysis.

ITEM No.	CONSUMPTION	COST.
1	5000	0.65
2	31300	0.4
3	12800	0.4
4	5000	0.1
5	5500	0.3
6	22650	0.5
7	22650	0.05
8	14760	2.1
9	21520	6
10	9000	0.2
11	29940	0.5
12	24660	0.5

ABC analysis.

Item No.	Consumption.	Cost.	AUV.	Sorted AUV	Proportional usage value.	Cumulative.
1	5000	0.65	3250	129120	57.46	57.46
2	31300	0.4	12520	30996	13.79	71.25
3	12800	0.4	5120	14970	6.66	77.91
4	5000	0.1	500	12520	5.57	83.48
5	5500	0.3	1650	12330	5.49	88.97
6	22650	0.5	11325	11325	5.04	94.01
7	22650	0.05	11325		2.28	96.29
8	14760	2.1	30996	5120	1.45	97.74
9	21520	6	129120	3250	0.8	98.54
10	9000	0.2	1800	1800	0.73	99.27
11	29940	0.5	14970	1132.5	0.50	99.77
12	24660	0.5	12330	500.	0.22	99.99
			$\Sigma AUV = 224713.5$			

Annual usage value, $AUV = \text{Annual demand} \times \text{cost}$
 $= \text{Consumption} \times \text{cost}$.

Proportional usage value $= \frac{AUV}{\text{Total}} = PUV$.

Cumulative = 1st value, 2nd value + 1st value, 1st + 2nd + 3rd value ... of PUV

FSN analysis.

Items No.	Consumption	Cost	Sorted consumption.	Proportional usage value	Cumulative
1	5000	0.65	31300	15.30	15.30
2	31300	0.4	29940	14.62	29.92
3	12800	0.4	24660	12.04	41.96
4	5000	0.1	22650	11.06	53.02
5	5500	0.3	22650	11.06	64.08
6	22650	0.5	21520	10.51	74.59
7	22650	0.05	14760	7.21	81.8
8	14760	2.1	12800	6.25	88.05
9	21520	6	9000	4.40	92.45
10	9000	0.2	5500	2.70	95.15
11	29940	0.5	5000	2.44	97.59
12.	24660	0.5	5000	2.44	100.03

$$\Sigma x = 204780$$

HML analysis.

H : High

M : Medium

L : Low

Item No.	Consumption.	Cost.	Sorted cost.	Proportional usage value	cumulative.
1	5000	0.65	6	51.3	51.3
2	31300	0.4	2.1	18	69.3
3	12800	0.4	0.65	5.55	74.85
4	5000	0.1	0.5	4.3	79.15
5	5500	0.3	0.5	4.3	83.45
6	22650	0.5	0.5	4.3	87.75
7	22650	0.05	0.4	3.42	91.17
8	14760	2.1	0.4	3.42	94.59
9	21520	6	0.3	2.56	97.15
10	9000	0.2	0.2	1.71	98.86
11	29940	0.5	0.1	0.85	99.71
12	24660	0.5	0.05	0.43	100.14

$$\Sigma = 11.7$$